Raja Marjieh

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Email: raja.marjieh@princeton.edu
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Google Scholar: https://tinyurl.com/4t9asvkb

Present Address
Science & Engineering Complex,
Harvard University, Boston, MA,
USA 02134

Academic Positions

Kempner Research Fellow, Harvard University — 2026-present
Kempner Institute for the Study of Natural and Artificial Intelligence

Education

PhD Program in Psychology, Princeton University — 2021-2026
Dissertation: Similarity in Natural and Artificial Minds
GPA: 4.0/4.0, Advisor: Prof. Thomas L. Griffiths

MSc in Physics, Technion, Israel Institute of Technology — 2016-2018
GPA: 96/100, Advisor: Prof. Amos Yarom

Joint BSc in Physics and Electrical Engineering, Technion — 2011-2016
GPA: 93.6/100, *cum laude*

Professional Experience

Student Researcher, Google Research — 2023

Lab Manager, Computational Auditory Perception Group, — 2019-2021
PI Dr. Nori Jacoby, Max Planck Institute for Empirical Aesthetics,
Frankfurt am Main, Germany

Intern, Decision Analytics Group - IBM Research Labs, Haifa — 2015

Research Assistant, Quantum and Ultrafast Devices Lab, — 2015-2016
Department of Electrical Engineering, Technion

Fellowships, Awards, and Honors

Kempner Research Fellowship, Harvard University	2026
Klarman Postdoctoral Fellowship, Cornell University (Declined)	2026
Miller Research Postdoctoral Fellowship, UC Berkeley (Declined)	2026
Princeton AI Lab Graduate Fellow	2025
Cognitive Science Society Travel Award	2023
Princeton Cognitive Science Program Graduate Funding Award	2021
Princeton First-Year Fellowship in Natural Sciences	2021
Katzir Travel Grant - Batsheva de Rothschild Fund and The Israel Academy of Sciences and Humanities	2018
The Clara Nevai and B. Bella Nevai Family Fellowship	2017
The Israel Council for Higher Education Fellowship for Outstanding Masters Students	2016
Norman and Barbara Seiden Family Prize for Multi-Disciplinary Undergraduate Projects	2015
Kasher Competition for Outstanding Student Projects, Honorable Mention	2015
Friedman Fund Undergraduate Fellowship	2014

Selected Publications

- **Marjieh, R.**, Harrison, P. M., Lee, H., Deligiannaki, F., & Jacoby, N. (2024). Timbral effects on consonance disentangle psychoacoustic mechanisms and suggest perceptual origins for musical scales. *Nature Communications*, 15(1), 1482.

- **Marjieh, R.**, Sucholutsky, I., van Rijn, P., Jacoby, N., & Griffiths, T. L. (2024). Large language models predict human sensory judgments across six modalities. *Scientific Reports*, 14(1), 21445.
- **Marjieh, R.**, Jacoby, N., Peterson, J. C., & Griffiths, T. L. (2024). The universal law of generalization holds for naturalistic stimuli. *Journal of Experimental Psychology: General*, 153(3), 573. (*Editor's Choice article*).
- **Marjieh, R**., van Rijn, P., Sucholutsky, I., Sumers, T. R., Lee, H., Griffiths, T. L., & Jacoby, N. (2023). Words are all you need? Capturing human sensory similarity with textual descriptors. *The Eleventh International Conference on Learning Representations* (*ICLR*, 2023).
- **Marjieh, R.**, Veselovsky, V., Griffiths, T. L., & Sucholutsky, I. (2026). What is a number, that a large language model may know it? *Transactions on Machine Learning Research* (*TMLR*, 2026).
- Harrison*, P., **Marjieh*, R.**, Adolfi, F., van Rijn, P., Anglada-Tort, M., Tchernichovski, O., ... & Jacoby, N. (2020). Gibbs sampling with people. *Advances in Neural Information Processing Systems*, 33, 10659-10671. (*NeurIPS*, 2020) (*Equal contribution; selected for oral).

Competitively Peer-reviewed Publications

1. **Marjieh, R.**, Veselovsky, V., Griffiths, T. L., & Sucholutsky, I. (2026). What is a number, that a large language model may know it? *Transactions on Machine Learning Research.*
2. van Rijn, P., Sun, Y., Lee, H., **Marjieh, R.**, Sucholutsky, I., Lanzarini, F., Andre, E., & Jacoby, N. (2026). One Test, Many Tongues: Surveying Language Proficiency Across the Globe. *Proceedings of the National Academy of Sciences*, 123(13), e2420179123.
3. Nurisso, M., Fernando, J., Deshpande, R., Perotti, A., **Marjieh, R.**, Frankland, S. M., ..., Cohen, J. D., & Petri, G. (2026). Bound by semanticity: universal laws governing the generalization-identification tradeoff. *The Fourteenth International Conference on Learning Representations* (ICLR).
4. **Marjieh, R.**, van Rijn, P., Sucholutsky, I., Lee, H., Jacoby, N., & Griffiths, T. L. (2025). Characterizing the Large-Scale Structure of Multimodal Semantic Networks. *Cognitive Science*, 49(10), e70131.
5. Ku, A., Campbell, D., Bai, X., Geng, J., Liu, R., **Marjieh, R.**, ... & Griffiths, T. L. (2025). Levels of Analysis for Large Language Models. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 384(2320), 20250012.
6. Sucholutsky, I., Muttenthaler, L., Weller, A., Peng, A., Bobu, A., Kim, B., ..., **Marjieh, R.**, ... & Griffiths, T. L. (2025). Getting aligned on representational alignment. *Transactions on Machine Learning Research.*
7. **Marjieh, R.**, Harrison, P. M., Lee, H., Deligiannaki, F., & Jacoby, N. (2024). Timbral effects on consonance disentangle psychoacoustic mechanisms and suggest perceptual origins for musical scales. *Nature Communications*, 15(1), 1482.
8. Rathje, S., Mirea, D. M., Sucholutsky, I., **Marjieh, R.**, Robertson, C. E., & Van Bavel, J. (2024). GPT is an effective tool for multilingual psychological text analysis. *Proceedings of the National Academy of Sciences*, 121(34), e2308950121.
9. **Marjieh, R.**, Sucholutsky, I., van Rijn, P., Jacoby, N., & Griffiths, T. L. (2024). Large language models predict human sensory judgments across six modalities. *Scientific Reports*, 14(1), 21445.

10. **Marjieh, R.**, Jacoby, N., Peterson, J. C., & Griffiths, T. L. (2024). The universal law of generalization holds for naturalistic stimuli. *Journal of Experimental Psychology: General*, 153(3), 573. (*Editor's Choice article*).
11. Huang, D. M., Van Rijn, P., Sucholutsky, I., **Marjieh, R.**, & Jacoby, N. (2024). Characterizing Similarities and Divergences in Conversational Tones in Humans and LLMs by Sampling with People. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (ACL).
12. Tian, Y., Ravichander, A., Qin, L., Bras, R. L., **Marjieh, R.**, Peng, N., ... & Brahman, F. (2024). MacGyver: Are Large Language Models Creative Problem Solvers?. *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* (NAACL).
13. Sucholutsky, I., Battleday, R. M., Collins, K. M., **Marjieh, R.**, Peterson, J., Singh, P., ... & Griffiths, T. L. (2023). On the informativeness of supervision signals. *Uncertainty in Artificial Intelligence*, 2036-2046, PMLR (UAI).
14. **Marjieh, R.**, Sucholutsky, I., Langlois, T. A., Jacoby, N., & Griffiths, T. L. (2023). Analyzing diffusion as serial reproduction. *Proceedings of the 40th International Conference on Machine Learning*, PMLR 202:24005-24019 (ICML).
15. **Marjieh, R**., van Rijn, P., Sucholutsky, I., Sumers, T. R., Lee, H., Griffiths, T. L., & Jacoby, N. (2023). Words are all you need? Capturing human sensory similarity with textual descriptors. *The Eleventh International Conference on Learning Representations* (ICLR).
16. Kumar, S., Correa, C. G., Dasgupta, I., **Marjieh, R.**, Hu, M. Y., Hawkins, R., ... & Griffiths, T. L. (2022). Using natural language and program abstractions to instill human inductive biases in machines. *Advances in Neural Information Processing Systems*, 35, 167-180. *Outstanding paper award* (NeurIPS).
17. **Marjieh, R.**, Pinzani-Fokeeva, N., Tavor, B., & Yarom, A. (2022). Black Hole Supertranslations and Hydrodynamic Enstrophy. *Physical Review Letters*, 128(24), 241602.
18. **Marjieh, R.**, Pinzani-Fokeeva, N., & Yarom, A. (2022). Enstrophy from symmetry. *SciPost Physics*, 12(3), 085.
19. Harrison*, P., **Marjieh*, R.**, Adolfi, F., van Rijn, P., Anglada-Tort, M., Tchernichovski, O., ... & Jacoby, N. (2020). Gibbs sampling with people. *Advances in Neural Information Processing Systems*, 33, 10659-10671. (*Equal contribution) *Selected for oral presentation* (NeurIPS).
20. Jensen, K., **Marjieh, R.**, Pinzani-Fokeeva, N., & Yarom, A. (2019). An entropy current in superspace. *Journal of High Energy Physics*, 2019(1), 1-21.
21. Jensen, K., **Marjieh, R.**, Pinzani-Fokeeva, N., & Yarom, A. (2018). A panoply of Schwinger-Keldysh transport. *SciPost Physics*, 5(5), 053.
22. Panna, D., **Marjieh, R.**, Sabag, E., Rybak, L., Ribak, A., Kanigel, A., & Hayat, A. (2017). Linear-optical access to topological insulator surface states. *Applied Physics Letters*, 110(21).
23. Sabag, E., Bouscher, S., **Marjieh, R.**, & Hayat, A. (2017). Photonic Bell-state analysis based on semiconductor-superconductor structures. *Physical Review B*, 95(9), 094503.
24. Sabag, E., **Marjieh, R.**, & Hayat, A. (2016). Extreme multiphoton luminescence in GaAs. *Europhysics Letters*, 115(5), 57006.
25. **Marjieh, R.**, Sabag, E., & Hayat, A. (2016). Light amplification in semiconductor-superconductor structures. *New Journal of Physics*, 18(2), 023019. *Highlighted in*:

Donati, G., Light-matter interactions: Superconducting gain. *Nature Photonics* 10.4 (2016): 207-207.

Book Chapters

- Griffiths, T. L., Sanborn, A. N., **Marjieh, R.**, Langlois, T., Xu, J., & Jacoby, N. (2024). Estimating subjective probability distributions. In *Griffiths, T. L., Chater, N., & Tenenbaum, J. B. Bayesian models of cognition: Reverse-engineering the mind.* MIT Press.

Working Papers

- Shiiku*, S., **Marjieh*, R.**, Griffiths, T. L., Anglada-Tort, M., & Jacoby, N. (2026). Hybrid Human–AI Societies Balance Creativity and Diversity. *PsyArxiv* (R&R).
- Hu, H., **Marjieh, R.**, Collins, K. M., Li, C., Griffiths, T. L., Sucholutsky, I., & Jacoby, N. (2026). Why Human Guidance Matters in Collaborative Vibe Coding. arXiv preprint arXiv:2602.10473.
- Frankland*, S., **Marjieh*, R.**, Nurisso*, M., Fluegemann, J., Webb, T., Petri, G., Lewis, R., & Cohen, J. D. (2026). Capacity Limits in Cognitive Processing Reflect the Curse of Generalization. *OSF Preprint.* (R&R)
- **Marjieh, R.**, Kumar, S., Campbell, D., Zhang, L., Snell, J., & Griffiths, T. L. (2026). Generative Similarity, Generalization, and Abstraction. *OSF Preprint* (under review).
- **Marjieh, R.**, Griffiths, T. L., & Jacoby, N. (2025). Composite Representations Lead to Multiple Geometrical Approximations to the Structure of Musical Pitch. *bioRxiv* (R&R).
- Budny, N., Ghods, K., Campbell, D., **Marjieh, R.**, Joshi, A., Kumar, S., ... & Griffiths, T. L. (2025). Visual serial processing deficits explain divergences in human and VLM reasoning. arXiv preprint arXiv:2509.25142.

Peer-reviewed Conference Proceedings

1. Cekinmez, J., Wu, A. J., **Marjieh, R.**, & Griffiths, T. L. (2026). Where did the ambiguity go? Examining how multimodal models interpret polysemous words. *COLM Workshop on Scientific Understanding of Foundation Models.*
2. Gaubert, L., Devaux, R., Çelen, E., **Marjieh, R.**, Mangalagiu, D., Jardin, A., & Jacoby, N. (2026). An Experimental Method to Study Opinion Diffusion in Human-AI Hybrid Societies. arXiv preprint arXiv:2605.09197. (CogSci 2026; Accepted).
3. Liu, Q. E., **Marjieh, R.**, Zhu, J. Q., Goldberg, A. E., & Griffiths, T. L. (2026). Parallelograms Strike Back: LLMs Generate Better Analogies than People. arXiv preprint arXiv:2603.19066. (CogSci 2026; Accepted).
4. Gautheron, L., **Marjieh, R.**, Conley, D. C., Frey, S., Rubin, H., Schneider, M. D., ... & Jacoby, N. (2026). Popularity Feedback Constrains Innovation in Cultural Markets. arXiv preprint arXiv:2602.09997. (CogSci 2026; Accepted).
5. Li, C., **Marjieh, R.**, Hu, H., Steyvers, M., Collins, K. M., Sucholutsky, I., & Jacoby, N. (2026). Human-AI Synergy Supports Collective Creative Search. arXiv preprint arXiv:2602.10001. (CogSci 2026; Accepted).
6. **Marjieh, R.**, Veselovsky, V., Griffiths, T. L. & Sucholutsky, I., (2025). What is a Number, That a Large Language Model May Know It? *NeurIPS Workshop on Interpreting Cognition in Deep Learning Models.*
7. **Marjieh, R.**, Anglada-Tort, M., Griffiths, T., & Jacoby, N. (2025). Characterizing the Interaction of Cultural Evolution Mechanisms in Experimental Social Networks. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 47.

8. Shiiku, S., **Marjieh, R.**, Anglada-Tort, M., & Jacoby, N. (2025). The Dynamics of Collective Creativity in Human-AI Societies. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 47.
9. Lee, H., Van Geert, E., Celen, E., **Marjieh, R.**, van Rijn, P., Park, M., & Jacoby, N. (2025). Visual and Auditory Aesthetic Preferences Across Cultures. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 47.
10. Liu, R., Yen, H., **Marjieh, R.**, Griffiths, T.L. & Krishna, R., (2025). Improving interpersonal communication by simulating audiences with large language models. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 47.
11. Liu, D., **Marjieh, R.**, Jacoby, N., & Hawkins, R. (2025). Two paths to variation in semantic judgments: How ambiguity and conceptual diversity drive individual differences in meaning. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 47.
12. **Marjieh, R.**, Gokhale, A., Bullo, F., & Griffiths, T. L. (2024). Task Allocation in Teams as a Multi-Armed Bandit. *ACM Proceedings of Collective Intelligence (CI '24)*.
13. **Marjieh, R.**, van Rijn, P., Sucholutsky, I., Lee, H., Griffiths, T., & Jacoby, N. (2024). A Rational Analysis of the Speech-to-Song Illusion. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46.
14. Urano, Y., **Marjieh, R.**, Griffiths, T., & Jacoby, N. (2024). The Influence of Social Information and Presentation Interface on Aesthetic Evaluations. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46.
15. Barretto, D., **Marjieh, R.**, & Griffiths, T. (2024). Reaching Consensus through Theory of Mind in Social Networks with Locally Distributed Interactions. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46.
16. Kumar*, S., **Marjieh*, R.**, Zhang, B., Campbell, D., Hu, M. Y., Bhatt, U., ... & Griffiths, T. L. (2024). Comparing Abstraction in Humans and Large Language Models Using Multimodal Serial Reproduction. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46. (*Equal contribution).
17. Niedermann, J., Sucholutsky, I., **Marjieh, R.**, Celen, E., Griffiths, T. L., Jacoby, N., & van Rijn, P. (2024). Studying the Effect of Globalization on Color Perception using Multilingual Online Recruitment and Large Language Models. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46.
18. **Marjieh, R.**, Sucholutsky, I., van Rijn, P., Jacoby, N., & Griffiths, T. L. (2023). What Language Reveals about Perception: Distilling Psychophysical Knowledge from Large Language Models. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 45.
19. van Rijn, P., Sun, Y., Lee, H., **Marjieh, R.**, Sucholutsky, I., Lanzarini, F., Andre, E., & Jacoby, N. (2023). Around the world in 60 words: A generative vocabulary test for online research. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 45.
20. **Marjieh, R.**, Sucholutsky, I., Sumers, T., Jacoby, N., & Griffiths, T. (2022). Predicting Human Similarity Judgments Using Large Language Models. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 44.

Teaching

Assistant in Instruction, Developmental Psychology, Princeton Department of Psychology, Level: UG — 2024

Head Assistant in Instruction, Computational Models of Cognition, Princeton Departments of Psychology and Computer Science, Level: UG/G — 2023

Head Assistant in Instruction, Probabilistic Models of Cognition, Princeton Departments of Psychology and Computer Science, Level: UG/G — 2022

Head Assistant in Instruction, Introduction to Solid State Physics, Technion Department of Electrical Engineering, Level: UG/G — 2015

Advising

- Burak Aydin (Graduate, Princeton Operations Research & Financial Engineering)
- Daniel Braga (Graduate, Princeton Computer Science)
- Yoko Urano (Undergraduate, Princeton Psychology)
- Rivka Mandelbaum (Undergraduate, Princeton Computer Science)
- Byron Zhang (Undergraduate, Princeton Computer Science)
- Daphne Barretto (Undergraduate, Princeton Computer Science)

Conference and Invited Talks

- Professional Skills in Behavioral Data Sciences, Nottingham Trent University, Nottingham, UK (February, 2026).
- Connecting Language and Perception in Natural and Artificial Minds, Department of Social & Decision Sciences, Carnegie Mellon University, Pittsburgh, PA (January, 2026).
- Similarity in Natural and Artificial Minds, Department of Psychology, Tufts University, Medford, MA (December, 2025).
- Similarity in Natural and Artificial Minds, Kempner Institute, Harvard University, Cambridge, MA (December, 2025).
- Aspects of Similarity in Natural and Artificial Minds, NYU Concats, New York University, New York, USA (October, 2025).
- Characterizing the Interaction of Cultural Evolution Mechanisms in Experimental Social Networks, CogSci, San Francisco, USA (July, 2025).
- Task Allocation in Teams as a Multi-Armed Bandit, ACM Collective Intelligence, Boston, USA (June, 2024).
- Integrating Human Decisions into Computer Algorithms, Workshop on Broader Adoption of Massive Online Experiments, CogSci, Sydney, Australia (July, 2023).
- What Language Reveals about Perception: Distilling Psychophysical Knowledge from Large Language Models, CogSci, Sydney, Australia (July, 2023).
- Analyzing Diffusion as Serial Reproduction, ICML (July, 2023).
- Words are all you need? Capturing human sensory similarity with textual descriptors, ICLR (May, 2023).
- New answers to old questions in semantic representation and music perception, Music Cognition Lab, Department of Music, Princeton (April, 2023).
- Predicting Human Similarity Judgments Using Large Language Models, CogSci, Toronto, Canada (July, 2022)
- Predicting Human Similarity Judgments Using Large Language Models, LNLS workshop, ACL, Dublin, Ireland (May, 2022).
- Characterizing the subjective pleasantness of tone combinations as a function of intervallic and spectral structure, ICMPC16-ESCOM11, Sheffield, UK (July, 2021).

- Revealing semantic representations through massive online experiments, CLaME, Max Planck - New York University Center for Language, Music and Emotion (September, 2020).
- Gibbs Sampling with People, Virtual Brains, Minds and Machines Summer Course (August 2020)
- Enstrophy and AdS4 Black Branes, Avant-Garde Methods for Quantum Field Theory and Gravity, Nazareth, Israel (2018)
- Semiconductor-Superconductor Two-Photon Amplifier, CLEO, San Jose, California, USA (2015)

Peer-review *CogSci*, *NeurIPS*, *PLOS ONE*, *PNAS Nexus*, *Nature Communications*.

Workshop Organization

- *Co-organizer*, AI for Scientific Discovery, Princeton (March, 2026).
- *Co-organizer*, Workshop on Networks and Cognition, Princeton (June, 2023).

Media Coverage

- Research featured in *New Scientist*, *Science News*, and the *Guardian*.
- Work on large language models and perception (Marjieh et al., *Scientific Reports*, 2024) featured in Agüera y Arcas, B. (2025). *What is intelligence? Lessons from AI about evolution, computing, and minds.* MIT Press.

Academic Community Service

Princeton Psychology: Served on two departmental committees. 2023-2025

Technion: Academic tutor at the Unit for the Advancement of Students. 2013-2015

Languages English, Arabic, Hebrew, Italian

Referees

Thomas L. Griffiths
Henry R. Luce Professor of Information Technology, Consciousness, and Culture
Departments of Psychology and Computer Science, Princeton University
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Nori Jacoby
Assistant Professor
Department of Psychology, Cornell University
Email: `kj338@cornell.edu`

Ofer Tchernichovski
Professor
Department of Psychology, Hunter College, City University of New York
Email: `otcherni@hunter.cuny.edu`

Jonathan D. Cohen
Robert Bendheim and Lynn Bendheim Thoman Professor
Departments of Neuroscience and Psychology, Princeton University
Email: `sp17@princeton.edu`